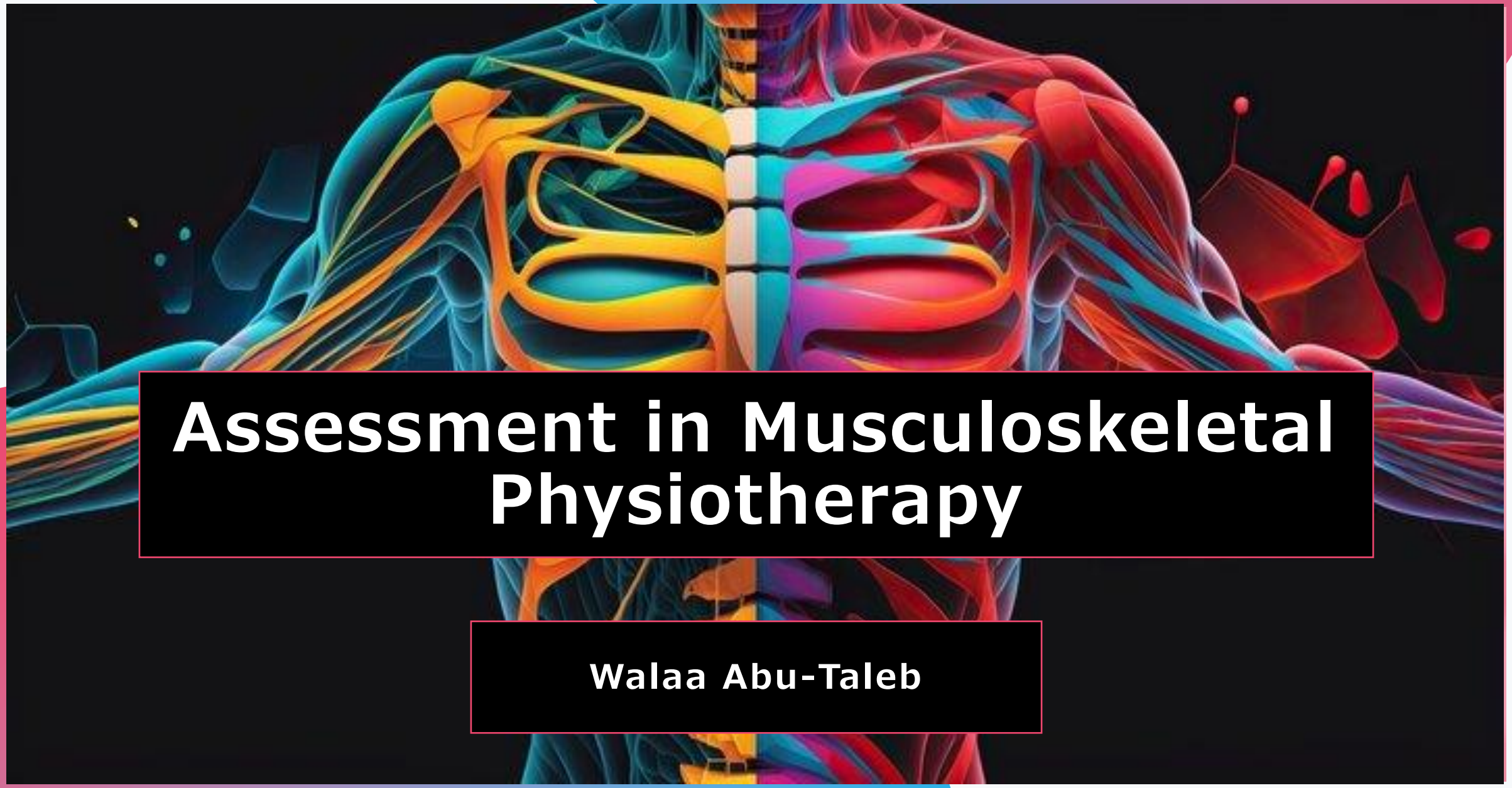


Assessment in Musculoskeletal Physiotherapy

Walaa Abu-Taleb

Learning outcomes

By the end of this Lecture, you should be able to:

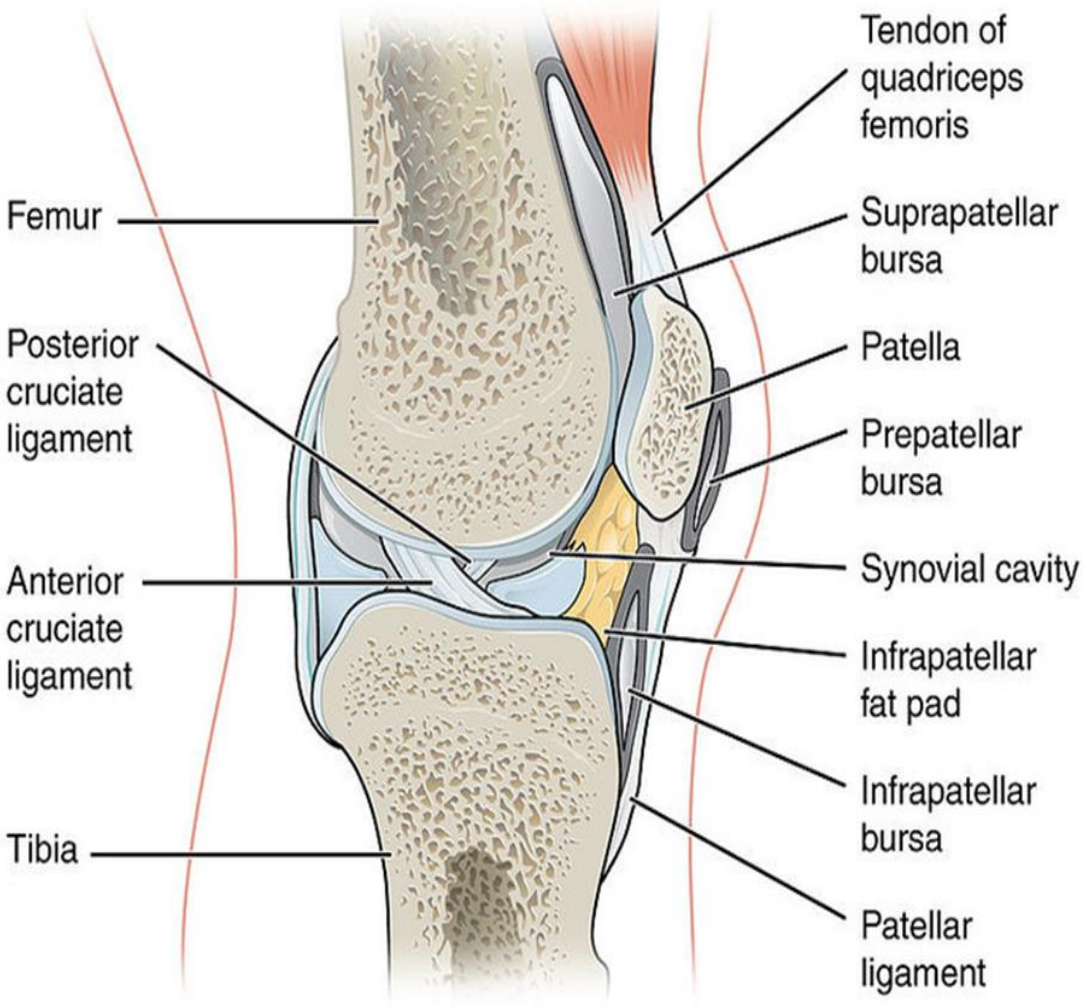
- Familiarise with the components of the musculoskeletal system in the human body.
- Understand the necessity of the biopsychosocial model in patient care.
- Differentiate between various causes of pain in the musculoskeletal system.
- Recognise the importance of the structural approach.
- understand the steps involved in conducting a comprehensive musculoskeletal assessment in order to apply them in case studies and clinical context.

The Musculoskeletal System

- The musculoskeletal system is an interconnected structure of bone, muscle, cartilage, tendons, and other tissues that give our bodies form, support, movement, and the ability to do work.

Functional Classification:

- The musculoskeletal system is made up of the bony skeleton and contractile and noncontractile soft tissues, including muscles, tendons, ligaments, joint capsules, articular cartilage, and nonarticular cartilage.



Movement System

Endocrine

Nervous

Cardiovascular

Pulmonary

Musculo-skeletal

Integumentary

Physical Examination

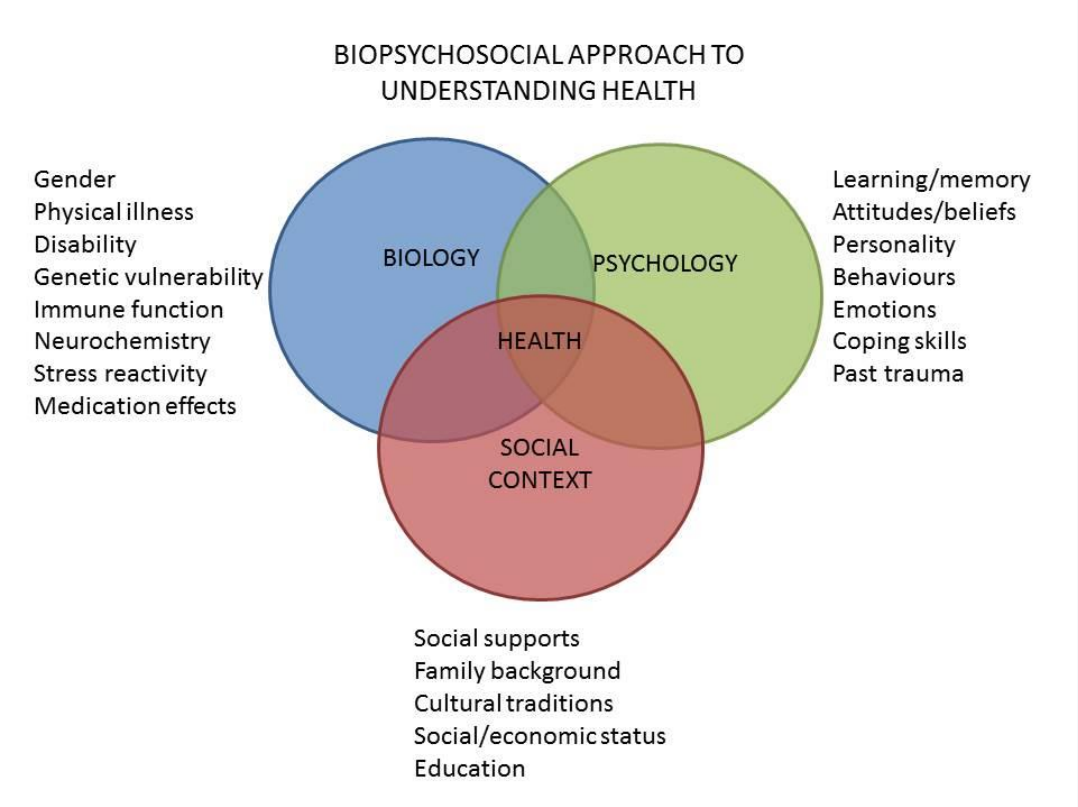
- Physical examination is the process of evaluating objective anatomic findings through the use of observation, palpation, percussion, and auscultation.
- The information obtained must be thoughtfully integrated with the patient's history and pathophysiology.
- It is a unique situation in which both patient and physician understand that the interaction is intended to be diagnostic and therapeutic.

Biomedical Model of Care

Biopsychosocial Model of Care

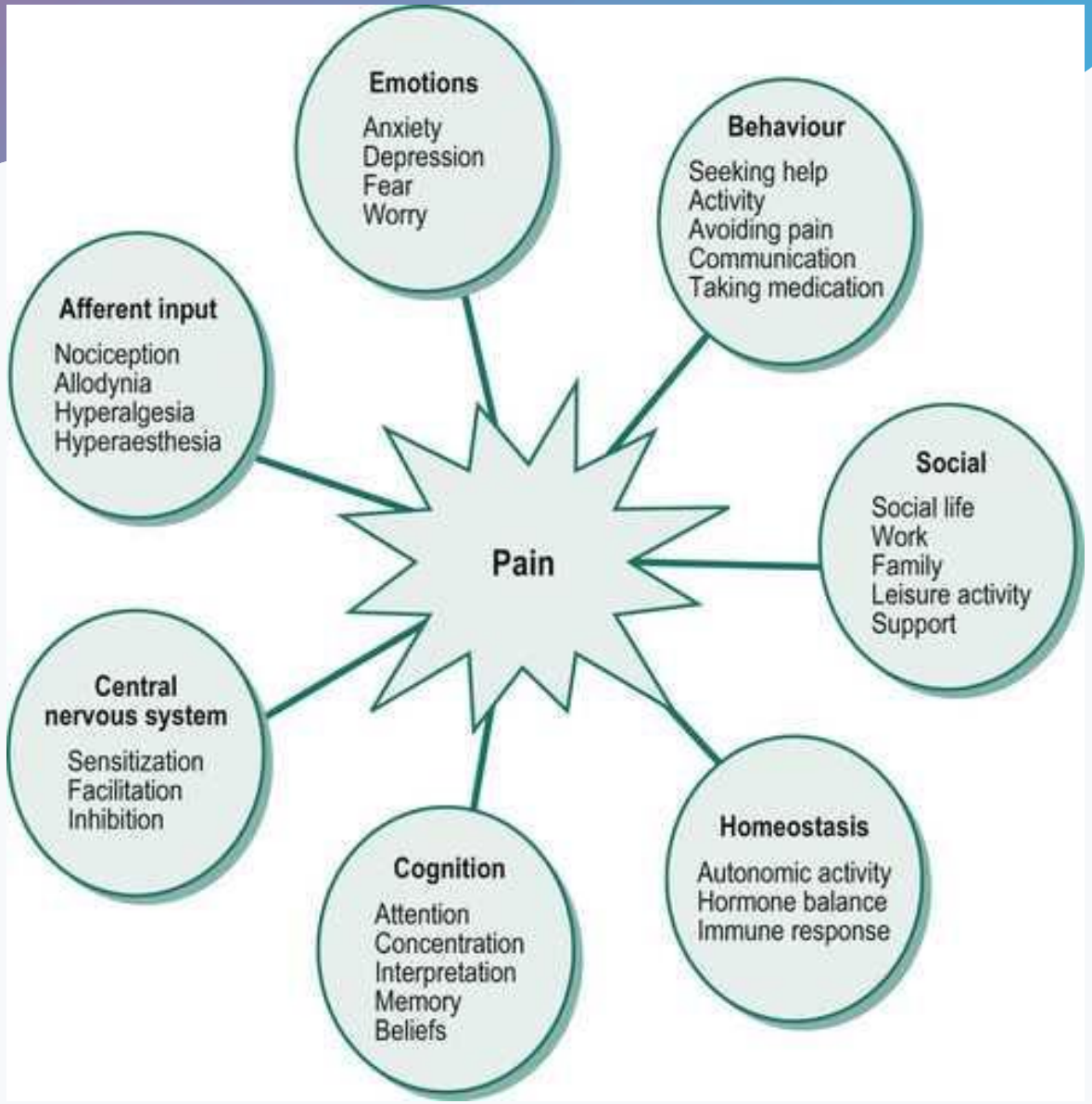
The Biopsychosocial Model

- **The model was developed by George L. Engel, a psychiatrist at the University of Rochester, and putatively discussed in an article published in 1977 in Science.**
- The biopsychosocial model is a general model positing that biological, psychological (which includes thoughts, emotions, and behaviors), and social (e.g., socioeconomical, socioenvironmental, and cultural) factors, all play a significant role in health and disease.
- It follows, that health and disease are best understood in terms of a combination of biological, psychological, and social factors rather than purely in biological terms.



Types of Pain

- 1- Nociceptive pain
- 2- Peripheral Neuropathic pain
- 3- Nociplastic pain (central sensitization)
- 4- Autonomic pain



Nociceptive Pain

Mechanical	Inflammatory	Ischaemic
Localized intermittent pain Predictable consistent response to mechanical stimuli, e.g. stretch, compression or movement No pain on waking but pain on rising Usually mild to moderate severity Responds to simple painkillers	Constant/varying pain Worsened rapidly by movement Latent pain Night pain and pain on waking High irritability and severity Movements limited by pain Responds to non-steroidal anti-inflammatory drugs (NSAIDs)	Usually intermittent Predictable pattern – aggravated by sustained postures and/or repetitive activities Eased by change of position or by cessation of a repetitive activity

Peripheral Neuropathic Pain

- Pain in the Neuroanatomical distribution, i.e. along a spinal segment or peripheral/cranial nerve pathway/course
- Typical descriptions include burning, sharp, shooting, electric shock-like, although neuropathic pain can manifest in many different ways:
    - allodynia (pain provoked by stimuli that are normally innocuous),
    - paraesthesia (abnormal sensation),
    - dysaesthesia (painful paraesthesia),
    - hypoaesthesia, hyperaesthesia, possibly a mixture of these
- Provoked by nerve stretch, compression or palpation. Possible associated hypoaesthesia or analgesia (partial or complete sensory loss), muscle weakness and autonomic changes
- Poor response to simple analgesia and anti-inflammatory medication. Response to passive treatment varies (Hansson & Kinnman 1996; Cook & van Griensven 2013)

Central Sensitization (Nociplastic)

- a patient may have pain as a consequence of **the change in the function of the somatosensory nervous system with no finding indicating nociceptive or neuropathic origin of pain.**

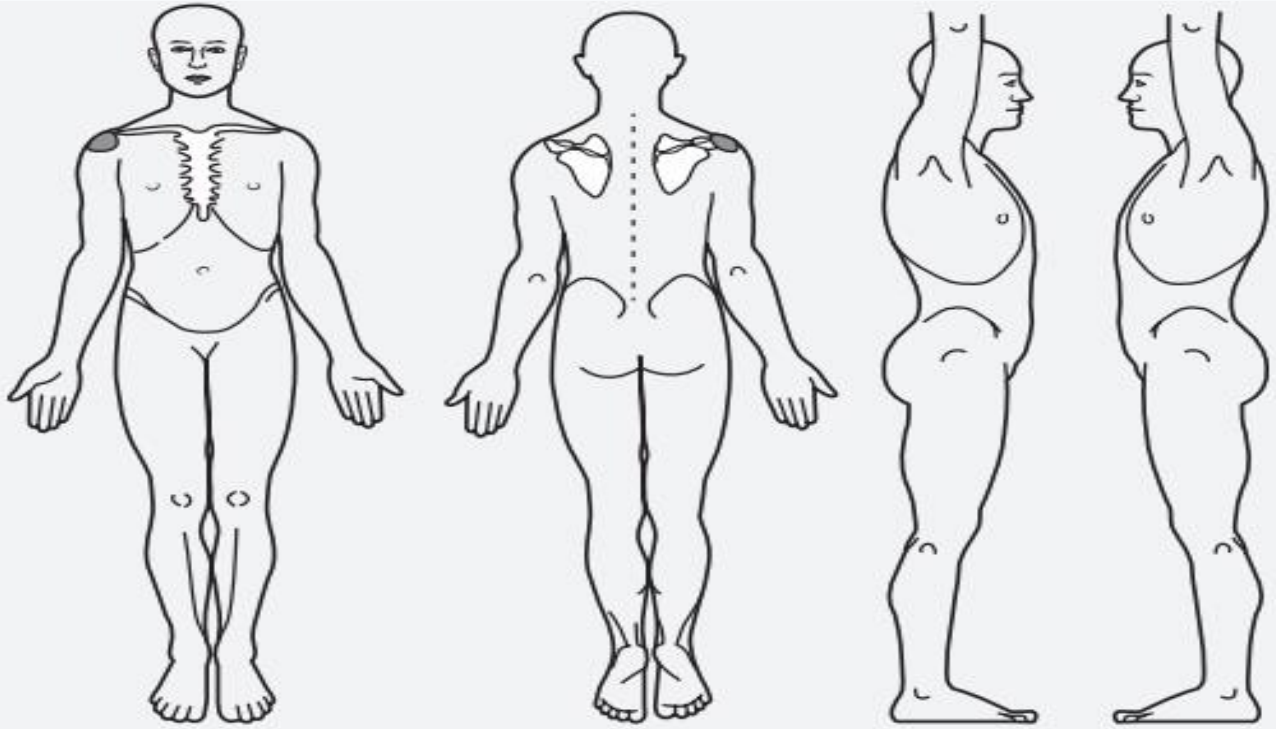
- Increased responsiveness of nociceptive neurons in the central nervous system to their normal or subthreshold afferent input (IASP 2015). This can develop as part of any persistent pain condition and be a consequence of ongoing nociceptive or neuropathic input: 'The following symptoms may suggest central sensitisation,but have to be interpreted with caution.'

    - Widespread, non-anatomical distribution Hyperalgesia (increased sensitivity to pain), allodynia evident
    - Inconsistent response to stimuli and tests
    - Patients have difficulty in locating and describing their pain
    - Pain seems to have 'a mind of its own'
    - Simple analgesics are ineffective
    - Unpredictable or failed response to passive treatments (Woolf 2012; van Griensven 2015)

Autonomic Pain

- It is the pain that is elicited by the activation of the sympathetic nervous system (SNS). SNS is responsible for Fight or flight response.
- This system can have an indirect impact on pain systems through the release of noradrenaline and cortisol affecting capillary response and indirectly cortisol inhibiting inflammation.
- Recognition of autonomic system arousal is important so it can be explained to the patient and managed using relaxation techniques, exercise and help with the origins of the patient's stress.
- May manifest locally as:
    - Discoloration
    - Dryness
    - Swelling
    - Sweating
    - Pilo erection

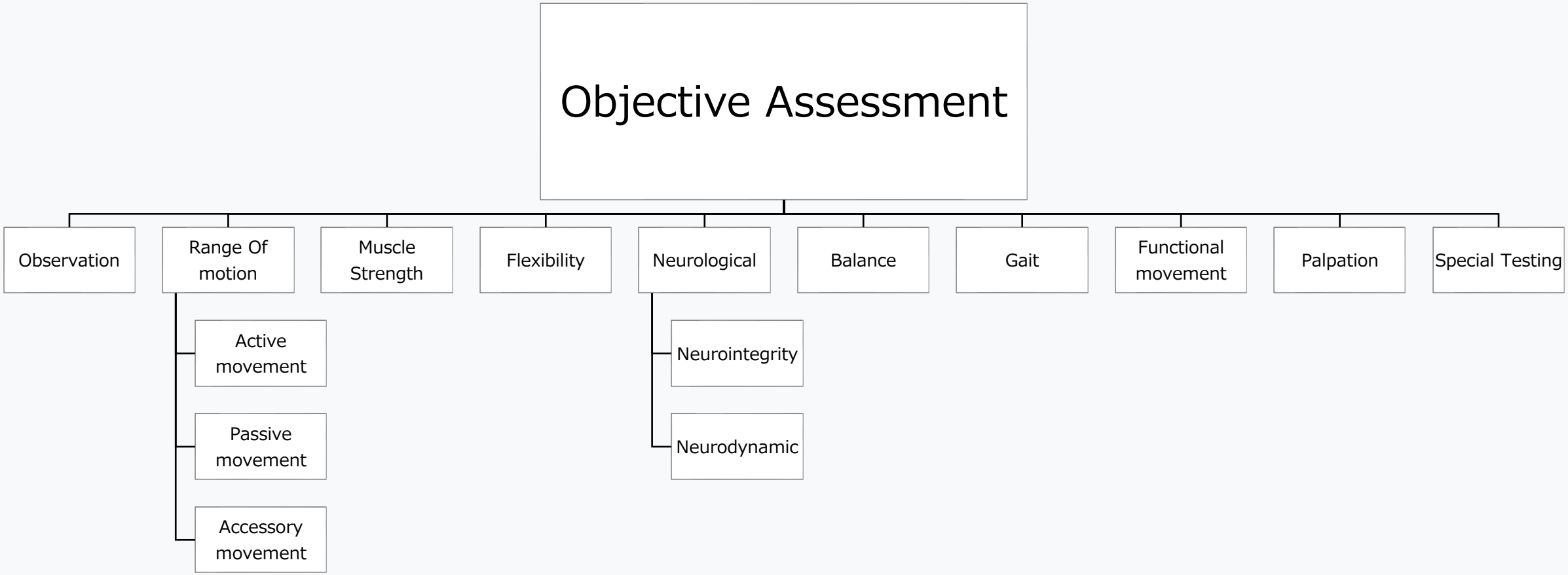
Body chart depicting an example of symptom location and the potential nociceptive, neuropathic and visceral sources that should be considered for that symptom area



Potential Local Nociceptive Sources	Potential Neuropathic Sources	Potential Nociceptive Sources of Somatic Referral	Potential Nociceptive Sources of Visceral Referral
• Glenohumeral periarticular (rotator interval structures, capsule and ligaments) • Glenohumeral intra-articular (glenoid labrum, biceps attachment, joint surface) • Subacromial space (rotator cuff, biceps, bursa, coraco-acromial ligament, acromion) • Acromioclavicular joint	• Axillary nerve • Suprascapular nerve • C3–C7 nerve roots	• Any C5/C6 motion segment structures (muscle, posterior intervertebral joint) • Any somatic structure sharing the C5–C6 innervation	• Visceral structures with common innervation to shoulder (e.g. phrenic nerve C3–C5 innervates diaphragm, pericardium, gallbladder, pancreas) • Visceral structures capable of irritating diaphragm (heart, spleen [L], kidneys, pancreas, gallbladder [R], liver [R])

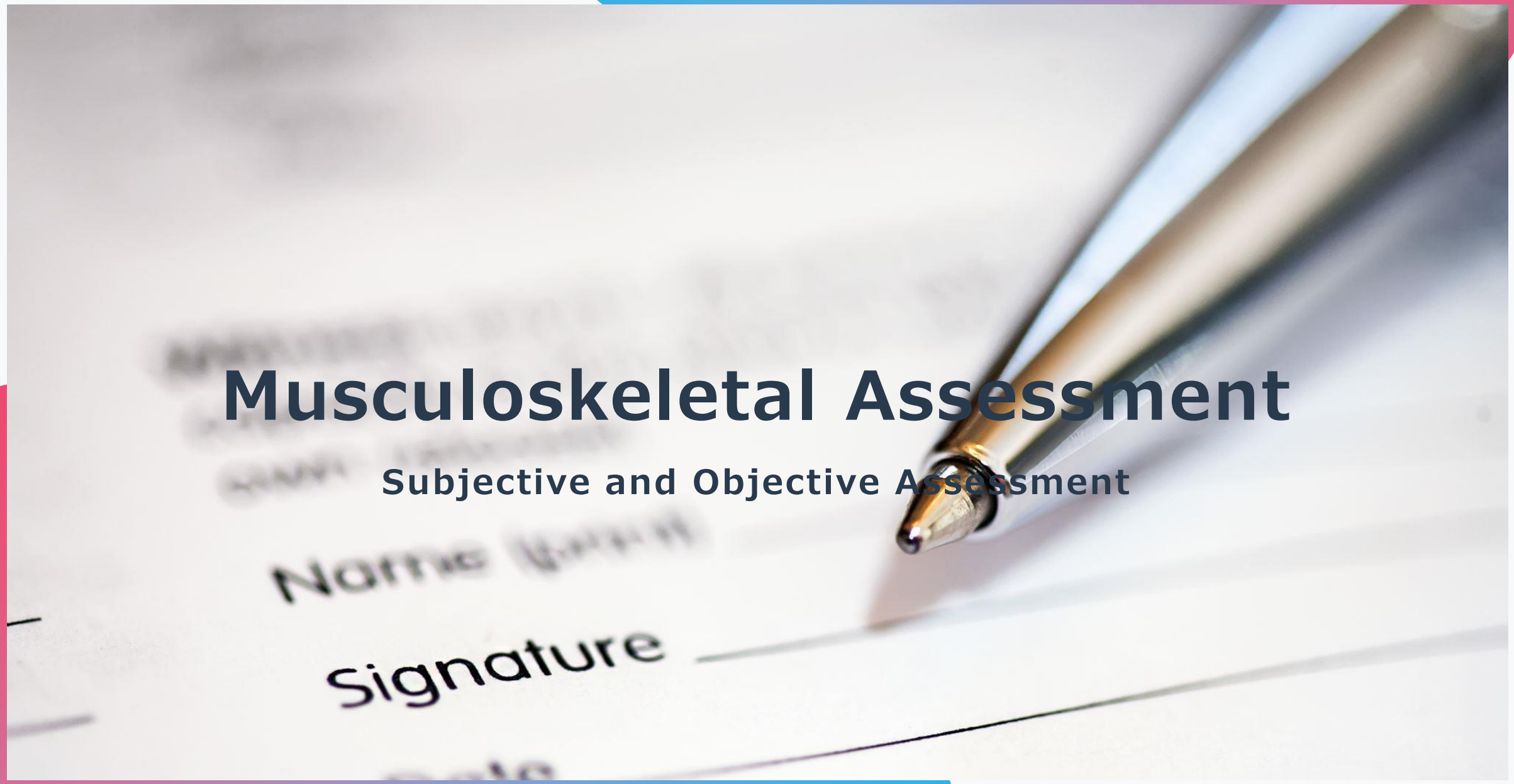
Musculoskeletal Examination

- **1- Patient Interview: (Subjective assessment, History taking, Goals, Patient perspective and expectations)**
- **2- Objective (Structural) Assessment:**



Musculoskeletal Assessment

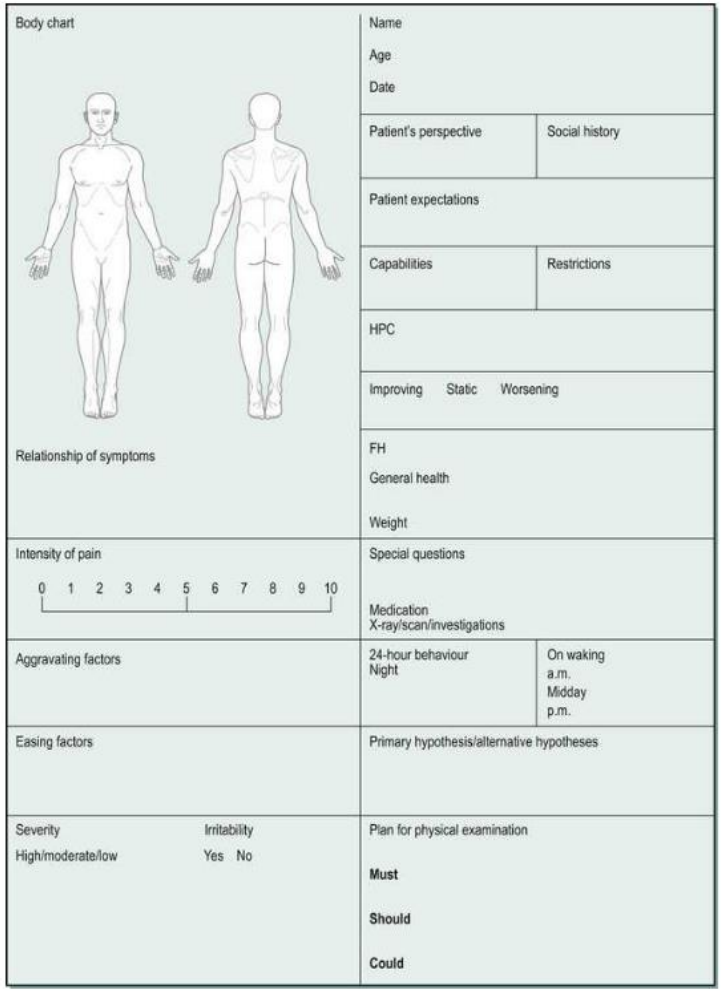
Subjective and Objective Assessment



Summary of Subjective Examination

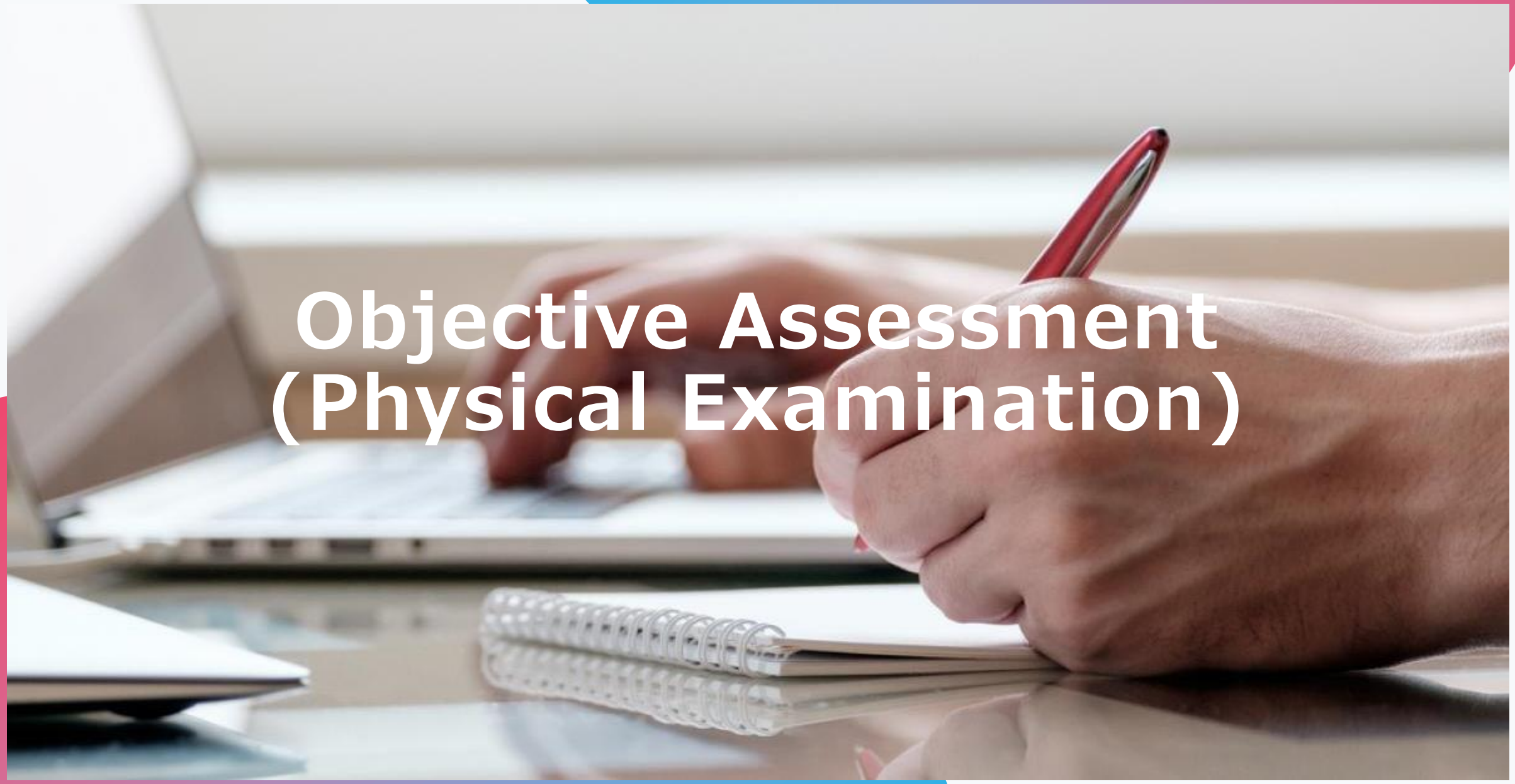
Information Gained	
Getting to know the patient	Patient expectations, beliefs, identifying their perspective
Social context	Age and gender, home and work situation, dependants and leisure activities
Body chart	Type and area of current symptoms, depth, quality, intensity, abnormal sensation, relationship of symptoms
Behaviour of symptoms	Aggravating factors, easing factors, severity and irritability of the condition, 24-hour behaviour, daily activities, stage of the condition
Special questions on family history	General health, drugs, steroids, anticoagulants, recent unexplained weight loss, rheumatoid arthritis, spinal cord or cauda equina symptoms, dizziness, recent radiographs
Past medical history	Relevant medical history, previous episodes, effect of previous treatment
History of present condition	History of each symptomatic area – how and when it started, how it has changed

Data from Nicholas, M., Linton, S., Watson, P., & Main, C. (2016) Early identification and management of psychosocial risk factors ('Yellow Flags') in patients with low back pain: a reappraisal. Physical Therapy (2016) 91(5):737–753.



Red flags

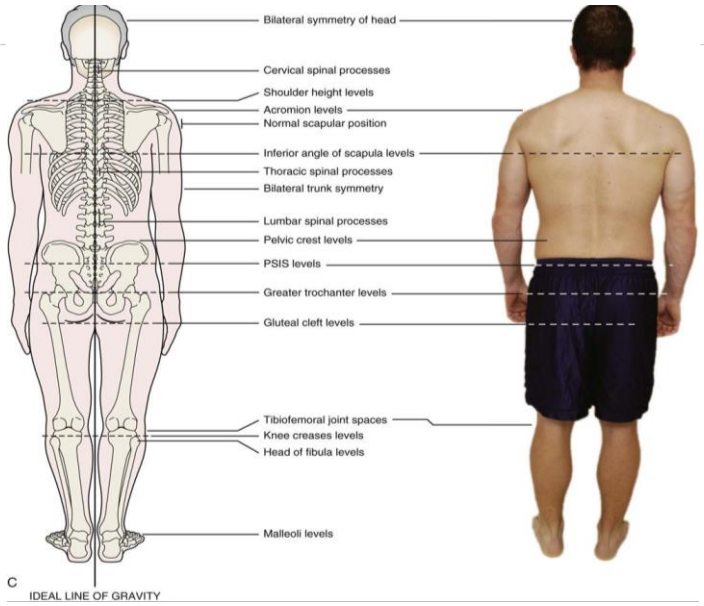
- Any recent violent trauma
- Unexpected/unplanned weight loss.
- Malaise or general weakness.
- Fever or night sweats.
- Constant or unremitting pain.
- Cancer
- TB
- Immunodeficient virus
- Cardiovascular disease
- Respiratory disease (Asthma)
- Epilepsy
- DM
- Thyroid illnesses
- Osteoporosis
- Osteoarthritis or RH arthritis
- Joint hypermobility
- Lumber stenosis
- Anxiety or depression
- Allergies
- **Case Specific red flags**
  - Dizziness- Cervical arterial artery late symptoms (5 Ds and 3 Ns)
  - Lumbopelvic symptoms (Cauda equina symptoms)



Objective Assessment (Physical Examination)

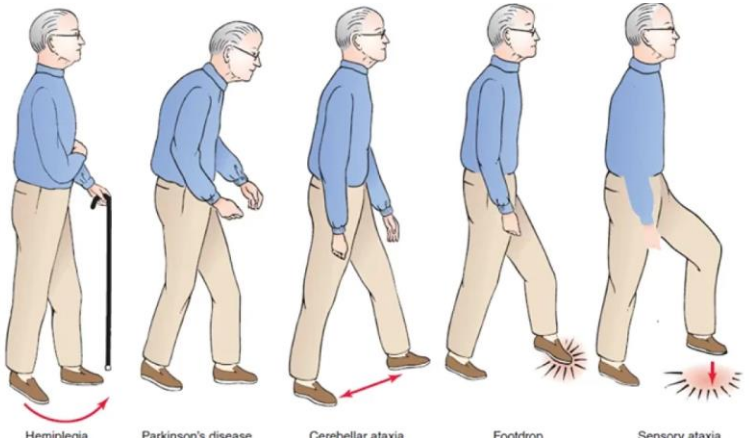
Observation

- Informal and formal observation of posture, muscle bulk and tone, soft tissues, gait, function and patient's response.
- Observation of Posture: standing , sitting, bony prominences, looking for symmetry and alignments
- Observation of structures: swelling, muscle atrophy, or hypertrophy, eyes, skin and nails, scars, traumas and contusions
- Observation of skin condition bruises, discoloration, scars or operation marks, dry skin – abrasions, ulcers, etc. – Brittle nails or clubbing, Hydration of skin – Abnormal hair growth – Other abnormal growths
- Signs of overt pain: guarding, bracing, rubbing, grimacing, sighing..etc
- Observation of gait and walking aids.
- Other observation, orientation level, speech, stability and coordination, mood and signs of discomfort









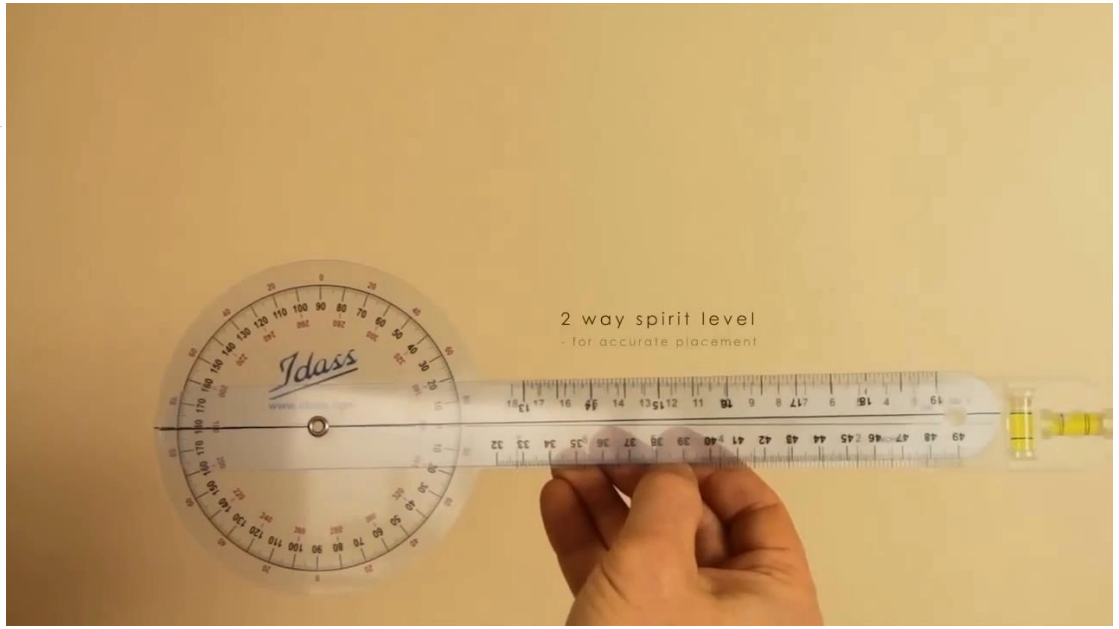


Range of Motion Assessment

- **1- Physiological movements:**

  A. **Active movements:** These are movements performed by the patient's voluntary muscular effort (e.g. flexion of the knee or abduction of the shoulder joint).

  B. **Passive movements**: These are movements performed by an external source, such as the physiotherapist or a pulley system.

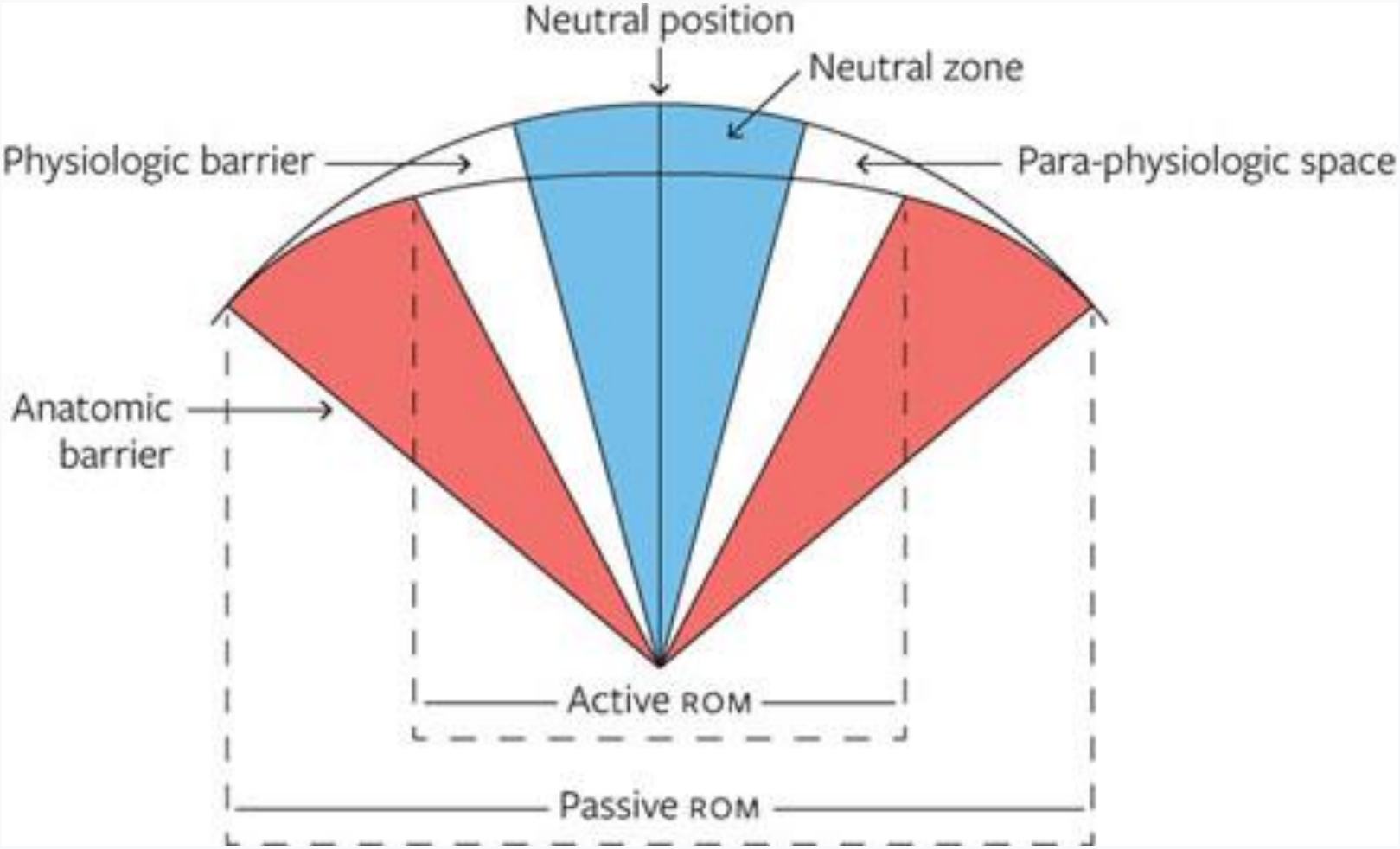




Active and Passive ROM



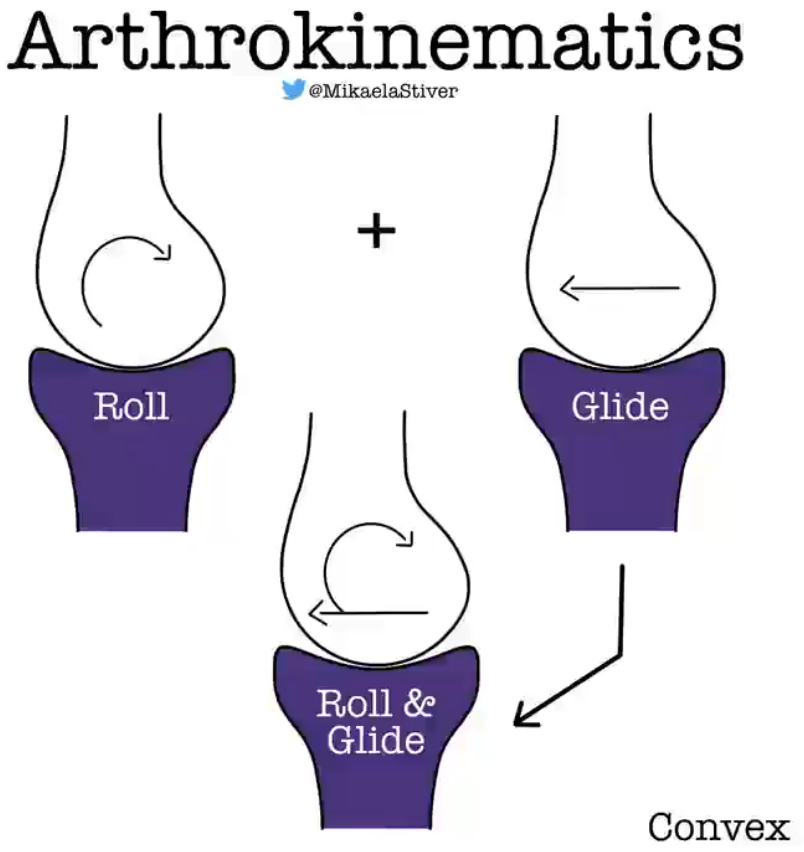
Active and Passive ROM and Anatomic barrier



	Joint	Normal Range of Motion (Degrees)
Shoulder	Flexion	0-180
	Extension	0-60
	Abduction	0-180
	Internal rotation	0-70
	External rotation	0-90†
Elbow	Flexion	0-150
Forearm	Pronation	0-80
	Supination	0-80
Wrist	Flexion	0-80
	Extension	0-70
Hip	Flexion	0-120
	Extension	0-30
	Abduction	0-45
	Adduction	0-30
	Internal rotation	0-45
	External rotation	0-45
Knee	Flexion	0-135
Ankle	Dorsiflexion	0-20
	Plantar flexion	0-50

Accessory movement

- **2- Passive Accessory Movements**
- A subtle movement that occurs between joint surface that cannot be seen by the eye or controlled actively. (e.g. they incorporate glide, roll or spin movements that occur in combination as part of normal physiological movements).
- An example of an accessory movement is an anterior–posterior glide at the knee joint.



Muscle Testing

- It is used to evaluate weakness and can be effective in differentiating true weakness from imbalance or poor endurance.
- The most accepted method of evaluating muscle strength is the Oxford Scale (AKA Medical Research Council Manual Muscle Testing scale). This method involves testing key muscles against the examiner's resistance and grading the patient's strength on a 0 to 5 scale accordingly.
- Modified Medical Research Council Scale

Grade	Definition
5	Normal strength
5−	Barely detectable weakness
4+	Same as grade 4, but muscle holds the joint against moderate to maximal resistance
4	Muscle holds the joint against a combination of gravity and moderate resistance
4−	Same as grade 4, but muscle holds the joint only against minimal resistance
3+	Muscle moves the joint fully against gravity and is capable of transient resistance, but collapses abruptly
3	Muscle cannot hold the joint against resistance, but moves the joint fully against gravity
3−	Muscle moves the joint against gravity, but not through full mechanical range of motion
2	Muscle moves the joint when gravity is eliminated
1	A flicker of movement is seen or felt in the muscle
0	No movement

Flexibility (Muscle Length) Assessment

Muscles Prone to Become Tight	Muscles Prone to Become Weak
Masseter, temporalis, digastric and suboccipital muscles, levator scapulae, rhomboid major and minor, upper trapezius, sternocleidomastoid, pectoralis major and minor scalenes, flexors of the upper limb, erector spinae (particularly thoracolumbar and cervical parts), quadratus lumborum, piriformis, tensor fasciae latae, rectus femoris, hamstrings, short hip adductors, tibialis posterior, gastrocnemius	Serratus anterior, middle and lower fibres of trapezius, deep neck flexors, mylohyoid, subscapularis, extensors of upper limb, gluteus maximus, medius and minimus, deep lumbar multifidus, iliopsoas, vastus medialis and lateralis, tibialis anterior and peronei

Common Muscle Patterns/Reaction of Muscles to Stress (Jull & Janda 1987; Janda 1994; Comerford & Mottram 2013)

Flexibility Ax

- the ability of a muscle crossing a joint or joints to lengthen, thus allowing the joint or joints to move through their full available range of motion.
- The muscle length assessment helps determine whether a muscle's length is normal, decreased or increased.
- This assessment can help identify if changes in muscle extensibility are contributing to a movement impairment and/or symptoms or if other structures are involved.
- To test muscle length, we must position the muscle so that the distance between its origin and insertion increases - i.e. we lengthen the muscle in the direction opposite to its action.

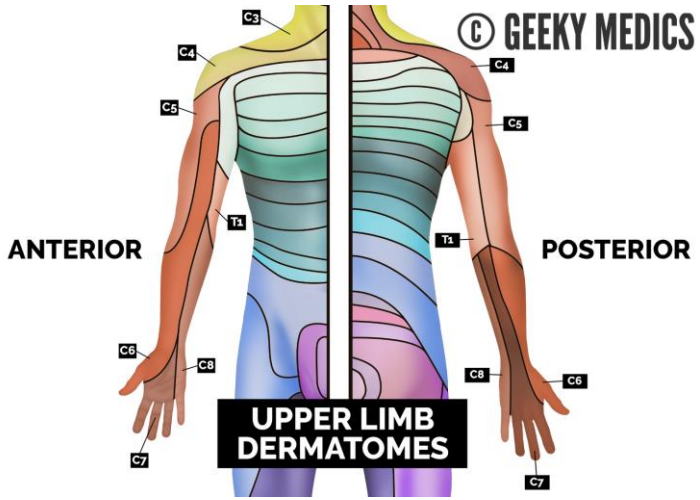


Neurological Assessment



NEURO-INTEGRITY AX

1. Dermatomes
2. Myotomes
3. Reflexes



NEURODYNAMIC AX

- **1- Upper Limbs:**
    - A. Median Nerve
    - B. Ulnar Nerve
    - C. Radial Nerve

2- Lower Limbs

- A. Sciatic Nerve
- B. Femoral Nerve

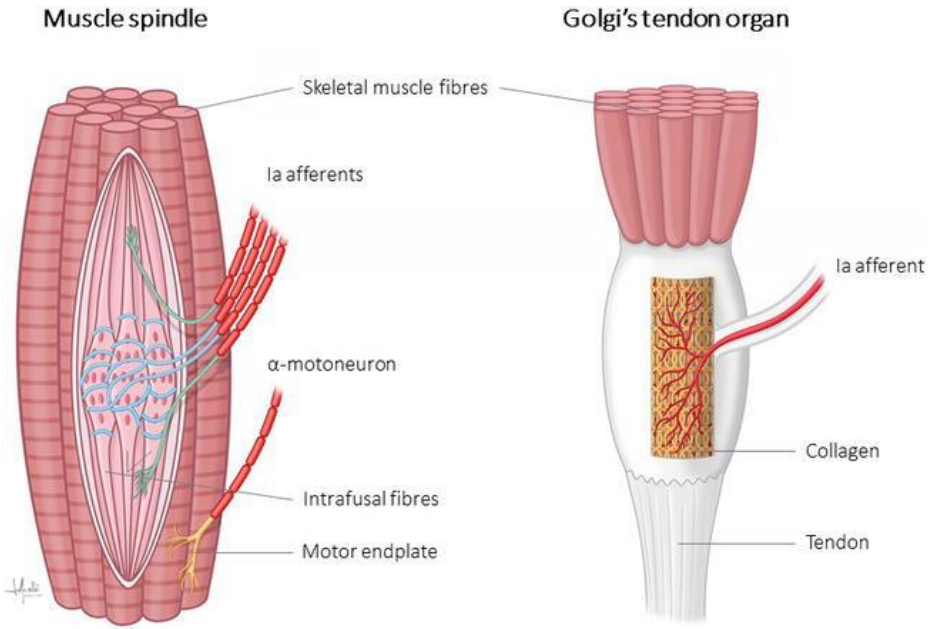
Proprioceptors assessment

Proprioception is the product of sensory information supplied by specialized nerve endings termed mechanoreceptors.

Proprioceptors and are found in muscle, tendon, joint and fascia, receptors in the skin can also contribute to proprioception.

Proprioception is critical for meaningful interactions with our surrounding environment.

They help with the planning of movements, sport performance, playing a musical instrument and ultimately helping us avoid an injury.



Gait Assessment

Human gait depends on a complex interplay of major parts of the nervous, musculoskeletal and cardiorespiratory systems.

Gait cycle: "The basic unit to characterise the way of walking, assuming that successive cycles will be reasonably similar if not the same."

This cycle is typically made up of 2 phases, namely **stance** and **swing** phase.

The cycle begins with one foot touching the ground and ends with the same foot touching the ground again.

Functional Ax

- The effect of MSK problems on the global functionality could be assessed by some tests:
- Sit to stand and stand to sit test
- Stair climbing test
- Reaching test
- Squatting test
- Standing on tip of toes

Palpation

- **Palpation** uses from light to firm pressure to identify and quantify the abnormalities of the musculoskeletal system. Normal or abnormal findings that could be elicited by observation and palpation include:

  - pain/tenderness,
  - trigger points.
  - symmetry/asymmetry
  - skin color and appearance, rash
  - lack of sweating hair abnormalities
  - warmth and heat
  - Swelling including effusions, nodules, and inflammatory findings like synovial and periarticular thickening
  - muscle atrophy, tone, contractures, and spasms
  - Crepitations
  - Joint deformities including spine like kyphosis and scoliosis.

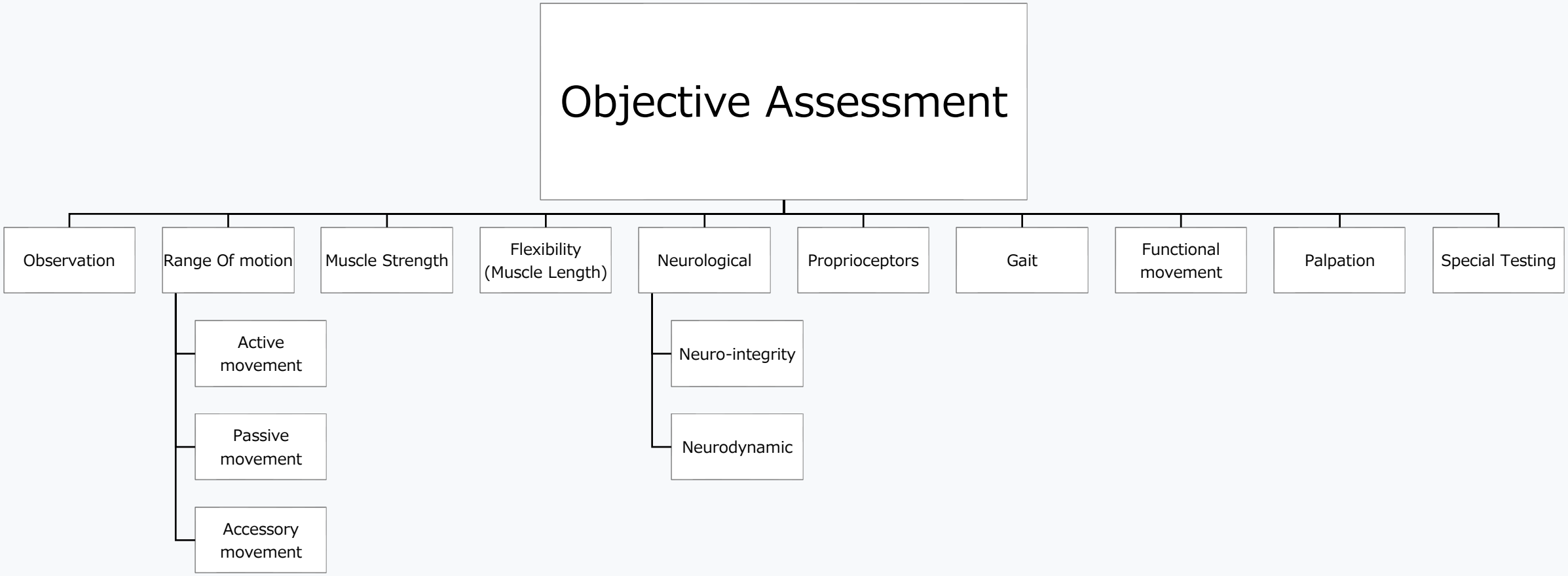
Special Testing

- These tests are conducted to examine for possible causes of the joint pain particularly soft tissue structures around the joint.

  - Special Tests (thestudentphysicaltherapist.com)

Musculoskeletal Examination

- **1- Patient Interview: (Subjective assessment, History taking, Goals, Patient perspective and expectations)**
- **2- Objective (Structural) Assessment:**



Questions

